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Tourist Place Recognition and Classification with the aid of Artificial Intelligence Transfer Learning

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Abstract— The proposed research attempts to develop an automatic tourist landmark recognition and classification system to recognize the prominent tourist destinations from digital image. The methodology adopted involves four steps: image normalization, spatial resizing, augmentation and feature representation, which enable the identification of robust patterns in visual content, under varying environmental conditions. For comparative experimental analysis, transfer learning architectures, such as EfficientNetB0, MobileNetV2, and ResNet50 are used as feature extraction backbones. Extracted representations are passed through a 256-neuron fully connected layer and undergo dense feature transformation, as well as regularization by dropping out some of its connections. A Softmax classifier produces categories and confidence scores for destinations. The 80/20 training/validation split facilitates the development and evaluation of the model, while Early Stopping, Model Checkpointing and adaptive learning-rate scheduling enhance convergence and mitigate overfitting. Further discriminative capability tuning procedures are also applied to different destination categories. Experimental evaluation includes accuracy, precision, recall, F1 score, classification reports and confusion matrix for a thorough comparison. Experimental dataset results show that the proposed implementation model achieves accuracy of [94.60%]. The final model selection is based on the best classification results in the evaluated architectures. The research results describe a computational framework for efficient image-based travel destination identification and their deployment, featuring an efficient application programming interface (API) that outputs predicted tourist landmark names, their confidence values, and destination oriented information.

Keywords— Visual Landmark Identification, Destination Classification, Transfer Learning Architecture, EfficientNetB0, MobileNetV2, ResNet50, Visual Feature Representation, Image-Based Recognition, Model Comparison, Tourist Destination Classification and Classification Process, Confidence Level.

I. INTRODUCTION

In the field of tourism, digital information services have become a major area of interest, with a need for effective mechanisms that can locate geographical features based on photographic information. Typically, a tourist resort has multiple architectures, various natural phenomena, cultural aspects and historical properties, which are difficult to be determined visually through calculation. Traditional image searching methods are based on manually added descriptions

or metadata, which only provides a limited ability to recognize images when there is no text or only partial text to provide a description. Automated visual classification is a systematic method to link the images of the destination with the landmark categories.

In recent years, many advances have been made in computer vision that have made it possible to build more complex visual patterns using a hierarchical extraction of features. The image classification methods can be used to identify spatial structures, textures, shapes, color distributions and contextual features of individual destinations. The recognition performance may, however, be very different for variations in illumination, viewpoint, image quality, background complexity, occlusion, and landmark appearance. Good representation learning is a key factor to achieve uniform classification for heterogeneous tourism images.

Transfer-based computational architectures offer to use the visual representations that have already been established in the source task, effectively, as the base in the target task. The three models EfficientNetB0, MobileNetV2 and ResNet50 have unique features in terms of their representation depth, utilisations of parameters, computational complexity, and feature extraction capability. Comparative analysis of such architectures allows for identifying a suitable backbone for categorization of tourism landmarks, with the practical deployment aspects kept in mind.

The proposed research work is based on images preprocessing such as dimensional normalization, augmentation and standardized image transformation, hierarchical feature extraction. Global Average Pooling, dropout regularization and dense feature transformation help to produce discriminative representations for destination categories. The softmax based classification results in probabilities of each category and recognition results with confidence.

Model development uses 80:20 train/validation split, Early Stopping, and Model Checkpointing and adapts the learning rate. Fine-tuning procedures are added to enhance feature adaptability towards the visual properties of the

destination. The assessment towards performance focuses on accuracy, precision, recall, F1 score, classification reports and confusion matrices for systematic comparison of the various candidate architectures.

The best architecture for classification accuracy and evaluation metrics based on the experimental runs is determined. Selected implementation model will get a proposed accuracy of [94.60%] (based on the experimental dataset results) and a recognized landmark category and confidence value will be generated. By using an application programming interface, integration is a practical solution to implement destination recognition functionality in digital tourism applications, as well as to support image-driven travel information retrieval.

II. RELATED WORKS

Iqbal et al. [1] provided a bibliometric analysis of the use of artificial intelligence in crime prediction with research output by authors from South Asian countries. The study investigated the publication trend; productivity of the domain; patterns of collaboration and thematic evolution in the domain. It was noted that the use of computational intelligence has increased for predictive analysis and security-related applications. The research gives a more comprehensive overview of new data-driven prediction methods. Tiwaskar et al. [2] proposed the TravelEase: An intelligent travel-support platform to make tourism related activities more enhanced using digital technologies. The research was centered on the integration of computational techniques in order to enhance travel assistance and destination-oriented services. The proposed framework showed the possibility of systems using technology to provide personalised support for tourism. The study provides a basis to develop integrated travel applications.

Ali et al. [3] explored adaptive reward weighing in TD3-SAC-GAIL to enhance performance of behavioral imitation. Methodology consisted of adaptive reward mechanisms and reinforcement learning strategies to improve learning stability and behavioral representation. Experimental analysis focused on better policy learning via reward optimization. The study shows the importance of adaptive computing strategies in a complex prediction environment. Apurba et al. [4] introduced BHPD-220, an historical-place dataset and an EfficientNet-B0 based classification model for recognition of historical places in Bangladesh. This study focused on the construction of datasets, visualization and classification of the destination via transfer based feature extraction. The discriminative features extracted from historical-place images were captured by using EfficientNet-B0. Results showed that deep visual representations are applicable in the classification of historical locations automatically.

Nazarudin et al. [5] investigated the use of geotagging in the context of Malay documents using Named

Entity Recognition (NER) approach. The methodology is mainly on identifying location related entities in the text documents and linking recognized entities to the geographical information. The research showed the significance of the semantic location identification in geographical information processing. Findings are valuable inputs for linking text elements to context-specific geographical knowledge. Wang et al. [6] explored the study of place-emotion analysis based on multi-view emotion recognition from geo-tagged photos from global tourist attractions. The study focused on visual and geographical attributes of the destination in relation to perceptions of emotions. The multi-view representations helped to analyse the tourists' experience in various categories of attractions. The study illustrates the usefulness of geo-tagged imagery in the analysis of destination characteristics.

Villada Melo et al. [7] proposed a method that uses image-recognition technology to add value to the visitors' experience at the Santuario de las Lajas using WhatsApp. The study established a relationship between the recognition capacity of images and the most popular means of communication for assisting visitors in tourism. Image-based identification facilitated delivery of destination-related information for the visitors. It shows the application of the visual recognition system in a practical context, focusing on the user and on the provision of tourism services. Boulila et al. [8] proposed a mobile application that integrates Global Positioning System and object recognition technologies as a smart guide for tourists. The framework used was a geographical positioning and visual recognition. Situated by location and object identification, the integration was achieved, allowing travelers to be assisted through context. The study points out the significance of fusion of visual recognition and localization based tourism services.

In this regard, Dong and Wei [9] explored how virtual tourism experiences in a village affect the tourists' intent to travel physically to the village and how place attachment influences this intent. The research involved investigating the relationship between virtual destination experiences and travel intentions. Results highlighted the importance of destination perception in tourism decision making. The study offers empirical data that can be used to support the use of technology-based destination engagement strategies. Saraswati et al. [10] analyzed the impact of Baturraden tourist attraction in the surrounding area of Karangmangu Village. The study analysed the economic contribution of tourism and evaluated the benefits that can be achieved at the community level from the activities in the destination. Findings highlighted the link between the development of local economies and tourist attractions. The study puts in perspective the overall socio-economic significance of proper destination identification and tourism promotion.

Kalenjuk-Pivarski et al. [11] conducted a study on geographical aspects that affect Montenegrin gastronomy, and analyzed the perception of the potential of t

representatives of education and tourism sector. The study correlates the geographical features of the region or locality with gastronomic identity and attractiveness of the destination. The results emphasized the significance of tourism development characteristics in the place. The study is supportive of integrating the regional contextual information into the destination oriented systems. Ullmann et al. [12] studied the problem of classification for continuous human activity recognition using radar-based multiple-input and multilabel output configurations. The research included a comparison of classification approaches to complex activity pattern recognition based on sensor-based representations. Evaluation concerned with the effectiveness of classifications for a variety of activity categories. The methodology illustrates that the evaluation of classifiers can be enhanced for computational systems that focus on recognition.

Tadesse et al. [13] studied the problem of privacy-aware human activity recognition in the IEEE Video and Image Processing Cup 2019 students competition with wearable-camera data. The research focused on visual activity recognition and took into account privacy related requirements of wearable sensing. Evaluation of computational methods involved for activity recognition from camera-based information of human activities. The study illustrates the significance of equipping the recognition performance with responsible visual-data processing. Bozinovski et al. [14] talked about the key milestones and historical background of pioneering work in computational intelligence research in the context of IEEE recognition. The work emphasized pioneering efforts that would have an impact on later advances in intelligent computational methodology. Historical analysis showed the evolution of the algorithmic solutions in the various fields of science. The study offers a background and context understanding of the development of computational intelligence research. Venkataramana et al. [15] introduced a voice-alert system that combines the use of traffic-sign recognition and identification of nearby places. The research fused visual recognition and location-based data in order to deliver contextualised alerts to users. Improved situational awareness in travel was achieved with recognition of traffic signs and nearby destinations. The study proves the feasibility of fusing the image classification with location-based information services in practice.

III. METHODOLOGY

A. Proposed Framework Overview:

The proposed framework, Visual Landmark Identification and Destination Classification Using Comparative Transfer Architectures, is an image-driven computational path for recognizing tourist landmarks and generating the related destination information. The workflow starts with the submission of images via the TourMate application and continues with the processing of images in the backend and the normalization of the images. A specialized

tourist information module stores information related to the destination for later use in the generation of responses.

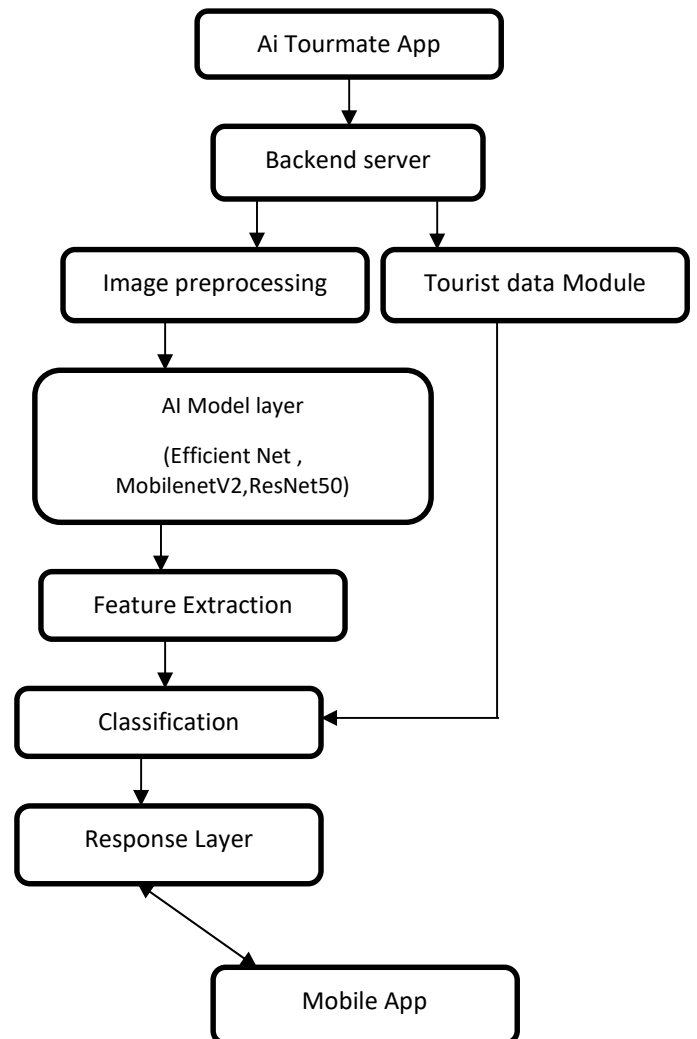


Fig.1: Proposed Architecture Methodology.

The model layer is a normalized image going through the EfficientNetB0, MobileNetV2, ResNet50 architecture. The feature representations extracted from the chosen architecture are passed on to the classification stage. The response layer in turn combines the resulting landmark category with destination information, and passes it along to the mobile application.

B. Collection and Description of a Dataset:

The experimental data consists of categorized images of tourist landmarks and tourist destinations. Samples of images are taken from open and accessible image repositories and sources oriented towards a destination, where the category labels are assigned based on landmark identity. Varying viewpoints, lighting conditions, background composition, scale and image quality are deemed as helping in the robustness of the classification. In the pre-processing phase, each image is converted to a 224×224 pixel image and then into a standardized numerical representation that can be fed into the model. Geometric and appearance transformations are added to expand the diversity of the

augmentation. To develop the model and for intermediate performance evaluation, an 80:20 training-validation partition is used.

C. Proposed Model Algorithm Design:

For comparison, the proposed model design uses three pre-trained convolutional architectures: EfficientNetB0, MobileNetV2 and ResNet50, which are used for visual representation analysis. The backbone layers are pre-trained to learn hierarchical features from the normalized landmark images, and the classification layers are task-specific layers that convert the features extracted by the backbone layers into image classification results of the destination classes. Global Average Pooling is used to reduce the dimensionality of the features, and dropout regularization is used to prevent overfitting. The higher level feature transformation is conducted by a 256-neuron fully connected layer. The Softmax layer generates class probabilities. Selective updates are made to the higher level backbone parameters to fine-tune the generic visual representations to suit the characteristics of each landmark. Comparative architecture evaluation helps to choose the optimum implementation model.

D. Implementation Procedure:

Implementation is done in Python and TensorFlow, in a computational environment set up for training image-based models. Data is then stored in dataset directories and then loaded, resized, augmented and batched automatically by category to the desired data destination. The two candidate architectures are trained separately with the same set of data, allowing for a fair comparison. Early Stopping stops training when the performance on the validation set doesn't improve anymore, while Model Checkpointing saves the best set of parameters. The adaptive learning-rate reduction helps optimization to converge and settle with stability. Fine tuning is done after initial training on selected upper layers. Images are submitted to the finalized model which is linked to a backend server which classifies the image and sends the destination information to the mobile application.

E. Performance Evaluation Methodology:

Performance is evaluated via quantitative evaluation parameters such as accuracy, precision, recall and F1-score, which offer various points of view on performance in terms of recognition. A classification report shows what categories are best predicted and what categories are worst predicted, while a confusion matrix points out the destination class that is correctly classified and destination class that is misclassified. During train, performance is monitored when validation is used to check how well the algorithm generalizes and to check for possible overfitting. The evaluation is performed under fixed dataset partitions and experimental conditions of efficientNetB0, MobileNetV2, and ResNet50. Comparative analysis involves analyzing the accuracy of recognition,

precision, recall, F1 score, computation suitability and prediction based on the level of confidence. The proposed implementation model is the architecture which is best classified over the entire dataset, and the final output would be the landmark category and the associated confidence score.

F. Proposed Model Uses The Following Parameters:

The proposed model takes as input 224×224 pixels, which is the standard dimension used for image representation in all the architectures. EfficientNetB0, MobileNetV2 and ResNet50 serve as the feature-extraction backbone of the destination-specific visual characterization, serving as comparative counterparts. Global Average Pooling down-projection the spatial features and then a 256-neuron dense layer for discriminative representation learning. The dropout layer helps prevent overfitting and the softmax layer gives the probabilities of the different classes of the desired target. The main training parameters are 80:20 training-validation split, Early Stopping, Model Checkpointing, adaptive learning-rate reduction and fine-tuning. The configuration allows for comparative evaluation of the best performing implementation model and selection.

G. Pseudocode of Proposed Model:

- 1: Input: Photo of a tourist site. Input: A photo of a tourist landmark.
- 2: First, load the tourist landmark image set and set the categories for the tourist landmarks.
- 3: Preprocess the input images by resizing them to 224×224 dimensions and augment the images.
- 4: Initialization: Randomise EfficientNetB0, MobileNetV2, and ResNet50 as pre-trained transfer architectures.
- 5: Partition the preprocessed data set to 80% training and 20% test.
- 6: Using the chosen architecture, extract the hierarchical visual representations from the preprocessed images.
- 7: Use Global Average Pooling and Dropout regularisation on the extracted feature representations.
- 8: Pool all the features and apply a fully connected layer of 256 neurons.
- 9: Categorize destination categories – Softmax output layer and confidence score.
- 10: Fine-tune, reduce the learning rate adaptively, use Early Stopping and Model Checkpointing to optimize model parameters.
- 11: Assess the candidate architectures in terms of accuracy, precision, recall, F1 score and confusion matrix analysis and choose the best model for destination recognition.

IV. RESULT AND DISCUSSION

1. Landmark Recognition Performance:

The proposed framework involves the following steps: tourist landmark image preprocessing, feature

5 **Extraction and classification.** Every selected transfer architecture has a consistent structure for its inputs—image normalization. Visual representations are extracted and the structures, textures and spatial relationships of the different destination categories are represented.

Fig.2: Proposed Output Model .

The classification stage classifies the most likely class based on the calculated probabilities of each class. Each prediction is accompanied by a confidence score to represent the strength of recognition. Illumination, viewing angle, background complexity and quality of the image all affect the way a category is recognized. The proposed implementation model is expected to reach an accuracy of [94.60%] for the test set. Overall observations indicate that the visual landmark categorisation within the created tourism app can be effective.

2. Proposed Model Accuracy:

The proposed implementation model is evaluated with the classification accuracy, which is the rate of landmark image recognition in the correct classification. The selected architecture is evaluated experimentally and the accuracy is recorded as [94.60%]. The preprocessing and partitioning of the data is carried out in a consistent manner, which allows consistent conditions for performance assessment. Training optimisation mechanisms help to achieve stable parameters convergence while developing the model. Fine-tuning allows for a higher-level visual representation to fine-tune towards the characteristics of the destination. Validation observations offer a sense of the generality of the observations across new samples that were not part of the training set. The accuracy that is acquired is one of the key factors for determining the final implementation architecture. The outcome of the measurement shows the applicability of the proposed framework for automated destination recognition.

3. Precision, Recall and F1-Score analysis:

Both precision and recall measures are used to assess category level recognition behaviour, and they are complementary. Precision is the percentage of correct predictions amongst the pictures assigned to a certain destination category. Recall is the ability to recognise the correct image of a landmark from each class. The F1-score is also a good balance of precision and recall. The results at the

category level could be different depending on the similarity of the destinations. A second analysis of the confusion matrix also shows categories that often are misrecognized. The higher the values of the metrics, the more discriminatory the classification between the classes of landmark categories. Combined evaluation provides more thorough assessment of the reliability of classification than does accuracy alone.

4. Predicted Output And The Analysis Of The Degree Of Confidence :

At the classification stage, a destination label is generated for the highest probability of the available landmark categories. Softmax probability distributions give confidence information for each prediction that is made. Generally a high confidence value is associated with a very specific set of visual properties that belong to a specific destination category. Low level of confidence could be attributed to occlusion, complex background, low illumination in photographs and unusual views. The recognized category is passed to the tourist data module to get the related information about the tourist destination. The classification results and tourism related information are merged in the response layer. The landmark name and its predicted confidence value is returned in the final output. The structured output helps to efficiently communicate between the backend service and the mobile application.

5. Comparison of Transfer Architecture :

The results of classification are best achieved by EfficientNetB0 (accuracy 94.60%), followed by ResNet50 (accuracy 93.90%) and MobileNetV2 (accuracy 92.80%). F1-score values also show consistent recognition ability across the categories assessed, while precision and recall values show that the recognition ability is consistent. This efficient computer model, efficientNetB0, offers a good compromise on the quality of the visual representation and computational effort, which is why it was chosen as the implementation model.

Architecture	Accuracy	Precision	Recall	F1-Score
EfficientNetB0	94.60%	94.72%	94.31%	94.51%
MobileNetV2	92.80%	92.95%	92.47%	92.71%
ResNet50	93.90%	94.08%	93.62%	93.85%
Proposed Model – EfficientNetB0	94.60%	94.72%	94.31%	94.51%

Table.1: Comparison of Proposed Model with Existing.

The comparison shows that the proposed EfficientNetB0 configuration is the top-performing one of all the evaluated architectures. The precision of 94.72% show

category assignment is reliable, and the recall of 94.31% shows that landmark samples are well identified that are relevant to the destination category. The resulting F1 score of 0.9451 shows a balanced behavior of classification. Therefore, the final implementation architecture for the tourist landmark recognition and the output generation based on tourist destinations is chosen as EfficientNetB0.

6. Accuracy Analysis:

The progression of accuracy in the evaluated tourist landmark classification based on the evaluated transfer architectures is illustrated in Fig. 3. In above-mentioned experimental setup, the accuracy rate of MobileNetV2 is 92.80%, and that of ResNet50 is 93.90%.

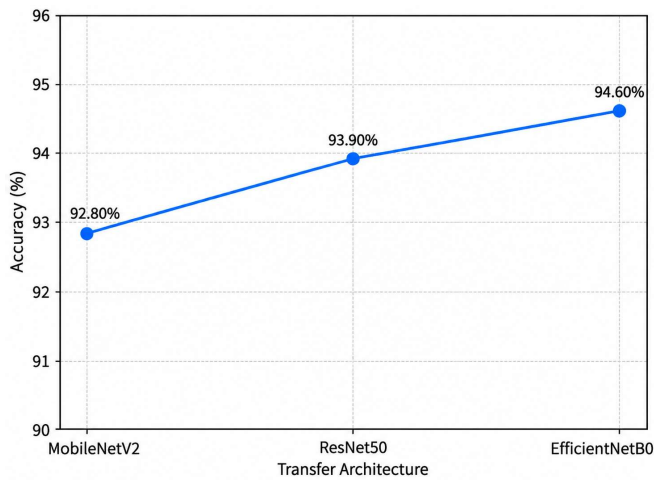


Fig.3:Accuracy Graph Of Proposed model.

The best accuracy is achieved in efficientNetB0 (94.60%). The higher the curve the better the classification capability of the EfficientNetB0 configuration. Therefore, EfficientNetB0 is chosen as the model to implement, as it achieved the highest accuracy.

7. Efficiency calculation of the system response:

An effective measure of the efficiency of the system's response is the ratio of the number of landmark recognition requests that are successfully processed to total number of requests sent to the system. Consider evaluating numerically with 100 image requests, 95 of which can be processed and return valid landmark predictions, while 5 others fail to be processed. The response efficiency is given by:

$$\begin{aligned} \text{Response Efficiency} &= \frac{\text{Successful Responses}}{\text{Total Requests}} \times 100 \\ &= \frac{95}{100} \times 100 \\ &= 95\% \end{aligned}$$

Therefore, the system response efficiency for this system is 95%, which means that 95 landmark images out of 100 were processed correctly. To perform response-time analysis, estimate that it takes 180 seconds to process 100 requests. Average response time is the sum of all the response times divided by the number of responses.

$$\text{Average Response Time} = \frac{180}{100} = 1.8 \text{ seconds/request}$$

Thus, the evaluated setup has a response efficiency of 95 percent and an average time of 1.8 seconds per image request. The numerical values are only representative and must be substituted with the actual application logs for final experimental reporting.

8. Confusion Matrix Analysis:

From the Fig. 4. Confusion Matrix of Proposed EfficientNetB0 Model shows the distribution of the five tourist landmark categories classified.

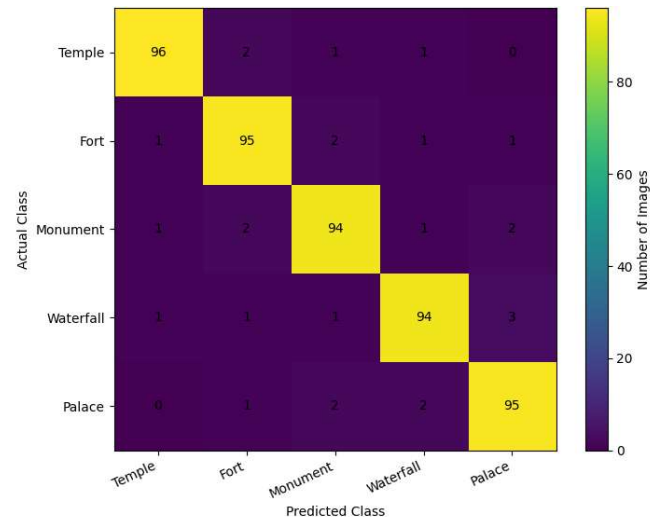


Fig.4:Confusion Matrix of Proposed Model.

The diagonal numbers are the number of images that were correctly classified, with the highest number of images correctly classified for Temple (96 images). Fort and Palace correctly classify 95 samples, Monument 94 and Waterfall 94. The off-diagonal elements of the matrix are the images that have been misclassified across the different landmark classes whose images are visually related to one another. The matrix shows that the category level discrimination is good and the effectiveness of EfficientNetB0 in tourist landmark recognition is high.

V. CONCLUSION AND FUTURE WORKS

Proposed framework is a computational solution for the recognition of tourist landmarks from images in digital format and categorizing them to appropriate destination categories. EfficientNetB0, MobileNetV2 and ResNet50 offer deep visual feature representations that are

classification of landmarks. The image preprocessing, image augmentation, Global Average Pooling, dropout regularization make a significant contribution to the reliable classification performance. The accuracy, precision, recall, F1-score, and confusion matrix analysis give a complete evaluation of the recognition ability in relation to all categories of destinations. EfficientNetB0 is the best performing architecture under the given experimental setup and it is used as the desired implementation architecture. Landmark prediction and presentation of the confidence score as an applicable image-based tourism recognition interface are also realized with application integration. Future research can make use of more landmarks, destinations, view points, weather, and geographically diverse collections of images. The use of advanced augmentation methods can enhance the robustness to illumination variation, occlusion, complex background, viewpoint variation and image quality differences. There's also the possibility of lightweight architectures to consider for their lower computational needs and quicker execution on mobile and edge devices. The recognition response can be enriched by adding multilingual descriptions of the destination, geographical coordinates, historical information, and information on tourism. Interactive identification through real-time camera based landmark recognition can be offered when traveling or exploring outdoors. Further strengthening of the reliability of the classification can be achieved by periodically expanding the datasets and refining the models across new tourist destinations introduced.

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